CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

Chapter 7: Ratio and Proportion

Previous year board questions, topic-wise: 13 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Properties of proportion

1. Use the properties of proportion to solve $\frac{6x^2+3x-5}{3x-5}=\frac{9x^2+2x+5}{2x+5}$, where $x \neq 0$.
ICSE 2025, 3 marks

2. Given $\frac{(a+b)^3}{(a-b)^3}=\frac{64}{27}$, find (a) $\frac{a+b}{a-b}$ (b) the ratio $a:b$, using the properties of proportion.
ICSE 2024, 4 marks

3. Use componendo and dividendo to solve $\frac{\sqrt{2x+2}+\sqrt{2x-1}}{\sqrt{2x+2}-\sqrt{2x-1}}=3$. ICSE 2023, 3 marks

4. Find $x:y$, given $\frac{x^2+2x}{2x+4}=\frac{y^2+3y}{3y+9}$. ICSE 2020, 3 marks

5. If $x=\frac{\sqrt{2a+1}+\sqrt{2a-1}}{\sqrt{2a+1}-\sqrt{2a-1}}$, prove that $x^2-4ax+1=0$. ICSE 2020, 4 marks

6. Use the properties of proportion to solve $\frac{\sqrt{5x}+\sqrt{2x-6}}{\sqrt{5x}-\sqrt{2x-6}}=4$. ICSE 2019, 3 marks

7. Use the properties of proportion to solve $\frac{2x+\sqrt{4x^2-1}}{2x-\sqrt{4x^2-1}}=4$, given that x is positive.
ICSE 2018, 3 marks

8. Given $\frac{7m+2n}{7m-2n}=\frac{5}{3}$, use the properties of proportion to find (a) $m:n$ (b) $\frac{m^2+n^2}{m^2-n^2}$.
ICSE 2017, 3 marks

2 Proportion

9. A train moves at a uniform speed. It covers 60 m in 2 s, 90 m in x s and y m in 5 s. Find x and y. (a) $x=4, y=150$ (b) $x=3, y=100$ (c) $x=4, y=100$ (d) $x=3, y=150$ ICSE 2024, 1 mark

10. What number must be added to each of 4, 6, 8 and 11 so that the four numbers are in proportion?
ICSE 2023, 3 marks

11. The numbers $K+3$, $K+2$, $3K-7$ and $2K-3$ are in proportion. Find K. ICSE 2019, 3 marks

3 Continued proportion

12. Find the mean proportional between 4 and 9. (a) 4 (b) 6 (c) 9 (d) 36 ICSE 2023, 1 mark

13. b is the mean proportional between a and c. Show that $\frac{a^4 + a^2b^2 + b^4}{b^4 + b^2c^2 + c^4} = \frac{a^2}{c^2}$. ICSE 2017, 3 marks

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